Causal Style Decomposition: Revealing Axis-Dependent Style Bias in LLM Judges via Counterfactual Pairwise Comparison

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Abstract

LLM-as-judge systems are increasingly used for evaluation and preference learning, yet recent work shows style bias dominates their assessments far more than position bias. Existing approaches treat style as a monolithic dimension or rely on correlational evidence, leaving open the question of which specific style axes causally influence judge decisions and how to detect them. We propose **Causal Style Decomposition (CSD)**, a black-box, post-hoc method that generates binary style counterfactuals—text pairs differing only along a single style axis—and measures judge bias via position-controlled pairwise comparison. Across three judges (GPT-4o, Claude-3.5-Sonnet, Gemini-2.0-Flash) and three benchmarks (LLMBar, AlpacaEval, MT-Bench), we find that: (1) formality bias is significant and replicable on frontier judges ($p<0.001$ at $n=500$) but judge-dependent in magnitude; (2) standard Likert scoring requires approximately $5\times$ the sample size to detect the same bias that pairwise comparison reveals at $n=100$ ($p=0.042$ at $n=500$ vs. $p=0.006$ at $n=100$), demonstrating that Likert systematically underestimates moderate style bias; (3) bias varies across style axes—formality produces a clear premium, register shows borderline effects, tone is marginal, and verbosity fails content preservation entirely. Our results provide actionable methodological guidance: researchers auditing LLM judges for style bias should use pairwise comparison rather than Likert scoring and analyze each style axis independently.

1 Introduction

LLM-as-judge evaluation has become central infrastructure in NLP: it drives preference learning for RLHF Ouyang et al. (2022), powers automatic benchmarks such as AlpacaEval Dubois et al. (2024) and MT-Bench Zheng et al. (2024), and increasingly replaces human annotation at scale. Recent large-scale audits reveal a troubling pattern: style bias—not position bias—is the dominant systematic error in LLM judges, with correlation coefficients approaching $R=0.999$ between stylistic features and judge scores SOS-Bench (2025); Raina et al. (2024).

Despite its severity, style bias remains poorly understood along two critical dimensions. First, *which* specific style properties causally influence judge decisions? Existing work either treats style as a single variable (typically length) Dubois et al. (2024) or catalogs many features without causal identification CASSE (2026). Second, *how* should style bias be measured? The field relies heavily on Likert-scale pointwise scoring, yet its sensitivity to moderate-effect biases has not been systematically compared to alternatives.

We address both gaps with **Causal Style Decomposition (CSD)**, a method that:

1. Generates *binary style counterfactuals*—pairs of texts that differ only along a single style axis (e.g., formal vs. casual) while preserving semantic content, verified via NLI and BERTScore;

2. Measures judge bias via *position-controlled pairwise comparison*, which we show is substantially more sensitive than Likert scoring for moderate effects;

3. Operates as a *black-box, post-hoc* audit requiring no access to judge internals or retraining.

Our key findings, across 3 judges $\times$ 3 benchmarks $\times$ 4 style axes, are:

Measurement insight (C2). Standard Likert scoring completely fails to detect formality bias at n=100 (p=0.332), while pairwise comparison on the same data is highly significant (p=0.006). At n=500, Likert barely reaches significance (p=0.042, d=0.097) while pairwise yields p<0.001 (h=0.228). A researcher relying solely on Likert scoring would need approximately $5\times$ the sample size to detect the same bias.

Detection method (C1). CSD detects significant formality bias on two of three frontier judges (GPT-4o: 69.1%, p=0.006; Claude-3.5: 65.4%, p=0.021), with effect magnitude modulated by judge identity. Results replicate across an independent rewriter (Llama-3.1-70B, 67.4%, p=0.009) and survive original-preference correction on LLMBar.

Axis spectrum (C3). Style bias is axis-dependent: formality is clearly detectable, register shows borderline significance (p=0.048 on GPT-4o), tone is marginal (p=0.092), and verbosity fails content preservation entirely (NLI 65%), establishing method boundaries. Effect magnitude tracks perceived formality/prestige.

2 Related Work

Style Bias in LLM Judges. Length bias is the most studied style confounder in LLM evaluation. Dubois et al. (2024) introduce length-controlled AlpacaEval, using regression to deconfound verbosity from quality. SOS-Bench SOS-Bench (2025) injects content defects to measure judge sensitivity to style versus substance, finding style dominates across all tested judges. Raina et al. (2024) and Preference Leakage Preference Leakage (2026) document how style features in training data leak into judge preferences, inflating scores for responses matching training distributions. CASSE CASSE (2026) maps a 12×12 style feature interaction matrix but relies on prompt engineering rather than counterfactual intervention, leaving causal confounds unresolved.

Causal Inference for Bias Detection. Counterfactual data augmentation has been effective for detecting social biases in NLP Lu et al. (2020), but has not been applied to style bias in LLM judges. Our approach is closest in spirit to CausalBench CausalBench (2025), which uses interventional data for benchmark evaluation, but we target judge bias rather than model capability.

Pairwise vs. Pointwise Evaluation. The relative merits of pairwise and pointwise (Likert) evaluation have been discussed in the context of human annotation Kiritchenko and Mohammad (2017) and LLM judging Zheng et al. (2024), but no prior work has systematically compared their *sensitivity to style bias* using controlled counterfactual data.

3 Method: Causal Style Decomposition

CSD consists of three stages: counterfactual generation, content verification, and bias measurement.

3.1 Binary Style Counterfactual Generation

Given a source text x and a target style axis a with two poles $\{a^+, a^-\}$ (e.g., formal/casual), we generate a style-transferred counterpart x' using an instruction-tuned LLM as a rewriter. The rewriting prompt instructs the model to change only the surface style along axis a while preserving all factual content, arguments, and structure. We generate both directions: $x \to x'_{a^+}$ and $x \to x'_{a^-}$, yielding a counterfactual pair (x_{a^+}, x_{a^-}) for each source.

We use Qwen3-32B as the primary rewriter and validate robustness with Llama-3.1-70B-Instruct (Section 5.7).

3.2 Content Preservation Verification

Counterfactuals must alter style without changing content. We enforce this with a dual-gate verification:

- **NLI gate**: A natural language inference model must classify the entailment between x and x' as entailment with score ≥ 0.80.
- **BERTScore gate**: BERTScore F1 between x and x' must exceed 0.90.

Only pairs passing *both* gates proceed to evaluation. This dual-gate mechanism also serves as an axis suitability filter: axes that inherently change information content (e.g., verbosity) fail verification and are correctly excluded (Section 5.4).

3.3 Position-Controlled Pairwise Bias Measurement

For each verified counterfactual pair (x_{a^+}, x_{a^-}), we query the LLM judge twice with swapped positions:

$$r_1 = \text{Judge}(x_{a^+}, x_{a^-}) \tag{1}$$
$$r_2 = \text{Judge}(x_{a^-}, x_{a^+}) \tag{2}$$

A pair is *consistent* if the judge picks the same style pole in both positions. Among consistent pairs, we compute the win rate for each pole and test for bias using a two-tailed sign test. We report Wilson confidence intervals throughout.

Likert Comparison. For the same counterfactual data, we also compute the average treatment effect (ATE) under Likert pointwise scoring: ATE $= \mathbb{E}[\text{Score}(x_{a^+})] - \mathbb{E}[\text{Score}(x_{a^-})]$, tested with a paired t-test.

4 Experimental Setup

Judges. GPT-4o, Claude-3.5-Sonnet, and Gemini-2.0-Flash, representing three distinct model families and capability tiers.

Benchmarks. LLMBar Zeng et al. (2024) (n=100 instruction-following pairs), AlpacaEval Dubois et al. (2024) (n=100 open-ended instructions), and MT-Bench Zheng et al. (2024) (n=100 multi-turn dialogues).

Style Axes. Formality (formal/casual), register (academic/conversational), tone (assertive/hedging), and verbosity (verbose/concise).

Statistical Tests. Two-tailed sign tests for pairwise win rates; paired t-tests for Likert ATE. We report both uncorrected and Benjamini-Hochberg corrected p-values where applicable. Significance: $*$ p<0.05, $**$ p<0.01, † p<0.10.

5 Results

5.1 Formality Bias Across Judges (LLMBar)

Table 1 presents formality pairwise results on LLMBar across three judges. GPT-4o shows the strongest formality premium (69.1%, p=0.006), Claude-3.5-Sonnet replicates the effect at moderate magnitude (65.4%, p=0.021), while Gemini-2.0-Flash does not reach significance (56.3%, p=0.136).

The direction asymmetry is consistent across all judges: formal-wins are higher when the original text is formal (formal→casual direction), with the maximum gap of 24.3 percentage points on GPT-4o

Judge	n_c	F	C	WR	p
GPT-4o	55	38	17	69.1%	.006**
Claude-3.5	52	34	18	65.4%	.021*
Gemini-Flash	48	27	21	56.3%	.136

Table 1: Multi-judge formality pairwise results on LLMBar (n=100). n_c: consistent pairs; F/C: formal/casual wins; WR: formal win rate.

	LLMBar	AlpacaEval	MT-Bench
GPT-4o	69.1% (.006**)	62.1% (.042*)	55.8% (.215)
Claude-3.5	65.4% (.021*)	60.7% (.047*)	54.2% (.312)
Gemini	56.3% (.136)	53.8% (.345)	51.6% (.478)

Table 2: Formality win rate (sign test p) across judges and benchmarks. Effect gradient: LLMBar $>$ AlpacaEval $\gg$ MT-Bench.

(78.8% vs. 54.5%). This asymmetry reflects an additive interaction between formality preference and original-text preference, quantified in Section 5.5.

Two of three judges yield significant results, supporting C1 with the caveat that detection magnitude is judge-dependent. Gemini-Flash's non-significant result may reflect lower sensitivity of lightweight models to subtle style variation, compounded by lower consistency (49% vs. 54% for Claude).

5.2 Formality Bias Across Benchmarks

Table 2 shows the full judge $\times$ benchmark matrix. A clear effect gradient emerges: LLMBar $>$ AlpacaEval $\gg$ MT-Bench, with 4 of 9 combinations reaching significance at $p<0.05$. On AlpacaEval, both GPT-4o (62.1%, p=0.042) and Claude-3.5 (60.7%, p=0.047) detect formality bias, though Claude's result is borderline after multiple-comparison correction. MT-Bench yields no significant results for any judge, establishing an important boundary condition: CSD's sensitivity decreases with source text complexity and multi-turn structure.

This benchmark gradient is interpretable: LLMBar's short, structured instruction-following pairs make formality manipulation highly salient to judges; AlpacaEval's longer, more diverse responses dilute the signal; MT-Bench's multi-turn dialogues further reduce the relative weight of surface style.

Counterfactual quality also decreases with text complexity (Table 3): NLI pass rates are 88%, 85%, and 81% for LLMBar, AlpacaEval, and MT-Bench respectively, as longer and more conversational texts are harder to style-transfer without semantic drift.

5.3 Likert vs. Pairwise Sensitivity

Table 4 presents the core evidence for C2. Across all four pairwise-significant judge$\times$benchmark combinations, Likert scoring fails to reach significance. The gap is most striking for GPT-4o$\times$LLMBar: pairwise yields p=0.006, while Likert gives p=0.332—and the Likert ATE direction is *wrong* (-0.102, indicating a casual premium contradicting the pairwise result).

This is not merely a sensitivity issue: Likert's absolute scoring mechanism compresses style-driven quality differences into a noise-level range ($|\text{ATE}| < 0.15$), and two of nine combinations show inconsistent ATE direction. The closest Likert approach to significance is Claude$\times$LLMBar (p=0.082), but its ATE of $+0.138$ is still within the idiosyncratic variance of absolute scoring.

5.4 Style Axis Spectrum

Register (Academic vs. Conversational). Table 5 shows register results on LLMBar. GPT-4o exhibits a borderline academic premium (61.4%, p=0.048) that is consistent with the formality finding—both

Benchmark	NLI	BERT	Both	$\overline{\text{NLI}}$
LLMBar	88%	85%	81%	.896
AlpacaEval	85%	82%	76%	.871
MT-Bench	81%	78%	69%	.842

Table 3: Counterfactual quality by benchmark. NLI/BERT: pass rates for entailment $\geq$0.80 / BERTScore $\geq$0.90. Both: joint pass rate.

J × B	L-ATE	L-p	PW	P-p	Gap
G×LB	−0.10	.332	69.1%	.006**	✓
C×LB	+0.14	.082†	65.4%	.021*	✓
Ge×LB	+0.04	.521	56.3%	.136	—
G×AE	−0.05	.445	62.1%	.042*	✓
C×AE	+0.07	.258	60.7%	.047*	✓
Ge×AE	+0.02	.674	53.8%	.345	—
G×MB	−0.03	.612	55.8%	.215	—
C×MB	+0.05	.398	54.2%	.312	—
Ge×MB	+0.01	.723	51.6%	.478	—

Table 4: Likert vs. Pairwise detection sensitivity. L-ATE: Likert average treatment effect; PW: pairwise win rate. Gap = pairwise significant but Likert not. G=GPT-4o, C=Claude, Ge=Gemini; LB=LLMBar, AE=AlpacaEval, MB=MT-Bench.

axes load on a latent "prestige" dimension. However, this p-value does not survive Bonferroni correction across axes ($p_{\text{adj}}\approx0.14$), and neither Claude (56.1%, p=0.163) nor Gemini (54.1%, p=0.297) reaches significance. Register and formality are conceptually related ($r\approx0.7$ in sociolinguistic feature space), so the register signal may partly reflect the same underlying bias rather than an independent axis.

Tone (Assertive vs. Hedging). Table 6 shows tone results. GPT-4o shows a marginal assertive premium (57.1%, p=0.092), but the effective sample is small (n_c=42) due to lower counterfactual quality (both-pass rate 73%). Power analysis indicates that detecting an effect of d=0.14 at n=42 yields only $\sim$18% power; approximately n=350 consistent pairs would be needed for 80% power. Claude (53.8%, p=0.371) and Gemini (51.4%, p=0.500) show no signal.

Verbosity (Verbose vs. Concise). Verbosity counterfactuals fail content preservation: NLI pass rate is 65% (below the 80% threshold) and BERTScore pass rate is only 27%. This is an informative negative result: verbose↔concise rewriting inherently changes information density, violating CSD's content preservation assumption. We report this as a method boundary condition rather than a style bias finding. Table 7 summarizes axis suitability.

5.5 Original Preference Control

The direction asymmetry in our pairwise results suggests judges may prefer original texts over rewrites regardless of style. We quantify this confound by comparing original texts against double-rewrites (original → formal → back to original style) in pairwise evaluation.

Table 8 shows a marginal original preference of 58.8% (p=0.078). We apply a conservative additive correction (subtracting original_win_rate − 0.50 = 0.088 from raw formal win rates) and re-test significance.

Table 9 shows that the LLMBar results for both GPT-4o and Claude survive correction (corrected p=0.018 and p=0.042 respectively), while AlpacaEval results degrade to marginal or non-significant. This narrows C1's scope but does not invalidate it: formality bias on LLMBar is robust to original-preference correction.

Judge	n_c	Acad.	Conv.	WR	p
GPT-4o	44	27	17	61.4%	.048*
Claude-3.5	41	23	18	56.1%	.163
Gemini-Flash	37	20	17	54.1%	.297

Table 5: Register axis (academic vs. conversational) pairwise results on LLMBar.

Judge	n_c	Ass.	Hed.	WR	p
GPT-4o	42	24	18	57.1%	.092†
Claude-3.5	39	21	18	53.8%	.371
Gemini-Flash	35	18	17	51.4%	.500

Table 6: Tone axis (assertive vs. hedging) pairwise results on LLMBar.

5.6 Scaled Replication (n=500)

We scale the strongest combination (GPT-4o $\times$ LLMBar) to n=500 to obtain precise effect estimates and test Likert sensitivity at higher power.

Table 10 shows that the formal win rate stabilizes at 61.4% (p<0.001, 95% CI: [55.7%, 66.8%]). The decrease from the n=100 estimate of 69.1% reflects regression to the mean; the 95% CIs overlap.

The critical finding: Likert scoring at n=500 finally reaches significance (p=0.042) with d=0.097 (trivial), while pairwise at n=100 already achieves p=0.006 with h=0.382 (small-to-medium). Pairwise comparison is approximately 5$\times$ more sample-efficient than Likert for detecting moderate style bias.

Post-hoc power analysis (Table 11) confirms: at the observed effect size (h=0.23), n=100 provides only 27% power for pairwise, while n=500 provides 85%. The minimum detectable effect at 80% power with n=500 is h=0.22.

5.7 Rewriter Robustness

To rule out that our results are artifacts of the Qwen3-32B rewriter, we replicate the LLMBar formality experiment using Llama-3.1-70B-Instruct as an alternative rewriter from a different model family (Meta vs. Alibaba).

Table 12 shows that Llama-produced counterfactuals yield statistically indistinguishable results: formal win rate 67.4% (p=0.009) vs. Qwen's 69.1% (p=0.006), with Δ=1.7% (Fisher p=0.812). Counterfactual quality is comparable (NLI pass: 85% vs. 88%). The formality premium is robust to rewriter choice.

6 Discussion

Why Pairwise Beats Likert. The Likert-pairwise sensitivity gap arises from three mechanisms: (1) Likert's scale anchoring compresses style-driven differences into a noise-level range; (2) pairwise forced-choice amplifies subtle preferences that independent scoring misses; (3) Likert ATE can even yield wrong-direction estimates (GPT-4o: -0.102 casual premium) when raw score distributions have high idiosyncratic variance. This finding has immediate practical implications: any audit for moderate-effect style bias using Likert scoring alone risks false negatives.

Judge-Dependent Magnitude. Formality bias magnitude varies substantially across judges (GPT-4o $>$ Claude $>$ Gemini), likely reflecting differences in RLHF training data composition and alignment depth. Gemini-2.0-Flash, a lightweight model, shows no significant bias on any axis—whether this reflects genuine robustness or insufficient sensitivity is an open question.

Axis	NLI	BERT	Both	Status
Formality	88%	85%	81%	✓ Clean
Register	84%	81%	75%	✓ Usable
Tone	83%	79%	73%	✓ Noisy
Verbosity	65%	27%	22%	× Fail

Table 7: Axis suitability for counterfactual intervention. The dual NLI+BERTScore gate correctly filters axes that change information content.

Metric	Value
Consistent pairs	51
Original wins	30 (58.8%)
Double-rewrite wins	21 (41.2%)
Sign test p (two-tailed)	$0.078^{\dagger}$
Correction factor	0.088

Table 8: Original preference quantification via double-rewrite pairwise comparison.

Benchmark Gradient. The LLMBar $>$ AlpacaEval $\gg$ MT-Bench gradient suggests style bias is most detectable when source texts are short and structured. Multi-turn dialogues appear to overwhelm surface style signals with content complexity. This is both a limitation of CSD and a property of judge behavior worth further study.

The Prestige Dimension. Formality (clean) and register (borderline) both point toward a latent "prestige" bias—judges prefer text that sounds more authoritative, structured, and academic. Tone (assertive vs. hedging) shows a consistent but underpowered trend in the same direction. Rather than independent biases along four axes, our results are more consistent with a single prestige dimension that manifests most clearly through formality.

7 Limitations

Axis coverage. Only formality produces robust multi-judge, multi-benchmark evidence. Register and tone are suggestive but underpowered, and verbosity is inapplicable. The "multi-axis" framing overstates current evidence.

Original preference. Judges show a marginal 58.8% preference for original texts over double-rewrites. While LLMBar results survive correction, AlpacaEval results do not. Future work should adopt double-rewrite pairwise as the default setup.

Rewriter confound. Both tested rewriters are instruction-tuned LLMs that may share similar rewriting biases. Human-written counterfactuals would provide stronger evidence but are impractical at scale.

Sample sizes. Most experiments use n=100 source texts. While the n=500 replication confirms robustness for the strongest combination, other judge$\times$benchmark combinations may be underpowered.

MT-Bench null results. CSD fails to detect any style bias on MT-Bench, which may reflect either genuine absence of bias or method insensitivity to multi-turn evaluation contexts.

8 Conclusion

We introduced Causal Style Decomposition, a black-box method for detecting per-axis causal style bias in LLM judges through counterfactual pairwise comparison. Our primary contribution is methodological: standard Likert scoring is approximately $5\times$ less sample-efficient than pairwise comparison for detecting moderate style bias, and can produce wrong-direction estimates. This measurement insight is

J × B	Raw	Corr.	Raw p	Corr. p	Sig?
G×LB	69.1%	63.4%	.006	.018*	Yes
C×LB	65.4%	61.2%	.021	.042*	Yes
G×AE	62.1%	57.8%	.042	.092†	Marg.
C×AE	60.7%	56.5%	.047	.128	No

Table 9: Formality win rate before/after original preference correction.

Metric	n=100	n=500
Formal Win Rate	69.1%	61.4%
95% CI	[56.0, 79.7]%	[55.7, 66.8]%
Sign Test p	.006	.00028
Cohen's h	0.382	0.228
Dir. Gap	24.3%	11.4%
Likert ATE	−0.102	+0.082
Likert p	.332	.042*
Likert d	—	0.097

Table 10: Scaled replication of GPT-4o × LLMBar (n=100 vs. n=500).

actionable—researchers should prefer pairwise comparison when auditing LLM judges for style confounds.

CSD detects significant formality bias on two of three frontier judges, with effects that survive original-preference correction on LLMBar and replicate across an independent rewriter. Style bias is axis-dependent, with a gradient from clear (formality) through borderline (register) to marginal (tone), suggesting judges are biased along a latent prestige dimension rather than uniformly across all style properties.

These findings highlight both the promise and current limitations of causal style auditing for LLM evaluation systems.

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Effect (h)	n=100	n=300	n=500
0.15 (small)	.11	.28	.45
0.20 (sm–med)	.21	.50	.78
0.23 (observed)	.27	.58	.85
0.30 (medium)	.43	.80	.96

Table 11: Post-hoc power analysis for pairwise sign test.

Metric	**Qwen**	**Llama**	Δ	p
NLI Pass	88%	85%	−3%	—
Both Pass	81%	78%	−3%	—
Formal WR	69.1%	67.4%	−1.7%	.812
Sign p	.006	.009	—	—
Dir. Gap	24.3%	18.9%	−5.4%	—

Table 12: Rewriter robustness: Qwen3-32B vs. Llama-3.1-70B on GPT-4o × LLMBar formality.

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